

Lab - 7

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25

1) Write a CUDA program to count the number of times a given word is repeated in a sentence (Use Atomic function).

```
#include "cuda_runtime.h" #include "device_launch_parameters.h" #include <stdio.h>
#include <stdlib.h> #include <string.h>
```

```
global void count_occ_of_word(char * str, char * word, int * space_indices, unsigned
int *d_count) { int i = threadIdx.x; int start, end; if (i == 0) { start = 0; end =
space_indices[0] - 1; } else { start = space_indices[i - 1] + 1; end = space_indices[i] - 1; }
```

```
int flag = 1, ind = 0;
for (int j = start; j <= end; j++) {
    if (word[ind++] != str[j]) {
        flag = 0;
        break;
    }
}
if (flag)
    atomicAdd(d_count, 1);

}
```

```
int main() { char str[100], word[100]; char * d_str, * d_word; int * d_space_indices;
unsigned int count = 0, *d_count, result;
```

```
printf("Enter a string ");
scanf("%[^\n]s", str);
fflush(stdin);
```

```

printf("Enter the word ");
scanf("%s", word);

int i = 0, space_count = 0, *space_indices = (int *) malloc(sizeof(int));
while (str[i] != '\0') {
    if (str[i] == ' ') {
        space_count++;
        space_indices = (int *) realloc(space_indices, space_count * sizeof(int));
        space_indices[space_count - 1] = i;
    }
    i++;
}
space_count++;
space_indices = (int *) realloc(space_indices, space_count * sizeof(int));
space_indices[space_count - 1] = strlen(str);

cudaMalloc((void **)&d_str, strlen(str) * sizeof(char));
cudaMalloc((void **)&d_word, strlen(word) * sizeof(char));
cudaMalloc((void **)&d_space_indices, space_count * sizeof(int));
cudaMalloc((void **)&d_count, sizeof(unsigned int));

cudaMemcpy(d_str, str, strlen(str) * sizeof(char), cudaMemcpyHostToDevice);
cudaMemcpy(d_word, word, strlen(word) * sizeof(char), cudaMemcpyHostToDevice);
cudaMemcpy(d_space_indices, space_indices, space_count * sizeof(int),
cudaMemcpyHostToDevice);
cudaMemcpy(d_count, &count, sizeof(unsigned int), cudaMemcpyHostToDevice);

count_occ_of_word<<<1, space_count>>>(d_str, d_word, d_space_indices, d_count);

cudaMemcpy(&result, d_count, sizeof(unsigned int), cudaMemcpyDeviceToHost);

printf("Total occurrences of word = %u\n", result);

cudaFree(d_str);
cudaFree(d_word);
cudaFree(d_space_indices);
cudaFree(d_count);
return 0;

}

```

OUTPUT

```
STUDENT@MIT-ICT-LAB5-25:~/230905213/week7$ nvcc q1.cu -o q1
STUDENT@MIT-ICT-LAB5-25:~/230905213/week7$ ./q1
Enter a string the cat and the dog saw the cat
Enter the word the
Total occurrences of word = 3
```

2) Write a CUDA program that reads a string S and produces string RS as: Input string S: PCAP Output string RS: PCAPPCAPCP Rule: Each work item copies the required number of characters from S into RS.

```
#include "cuda_runtime.h" #include "device_launch_parameters.h" #include <stdio.h>
#include <stdlib.h> #include <string.h>
```

```
global void pyramid_str(char * s, int l, char * rs) { int i = threadIdx.x; int start = (i * (2 * l +
((i - 1) * -1))) / 2; // Sum of AP for (int j = 0; j < l - i; j++) { rs[start + j] = s[j]; } }
```

```
int main() { char word[100]; char * d_word, * d_res; int wordLen;
```

```
printf("Enter a word ");
scanf("%s", word);
```

```
wordLen = strlen(word);
```

```
int totalchars = (wordLen * (wordLen + 1)) / 2; // Sum of natural numbers till n
```

```
char * res = (char *) malloc(sizeof(char) * (totalchars + 1));
```

```
cudaMalloc((void **)&d_word, wordLen * sizeof(char));
cudaMalloc((void **)&d_res, totalchars * sizeof(char));
```

```
cudaMemcpy(d_word, word, wordLen * sizeof(char), cudaMemcpyHostToDevice);
```

```
pyramid_str<<<1, wordLen>>>(d_word, wordLen, d_res);
```

```
cudaMemcpy(res, d_res, totalchars * sizeof(char), cudaMemcpyDeviceToHost);
```

```
res[totalchars] = '\0';
```

```
printf("Final str: %s\n", res);
```

```
cudaFree(d_word);  
cudaFree(d_res);  
return 0;
```

```
}
```

OUTPUT

```
STUDENT@MIT-ICT-LAB5-25:~/230905213/week7$ nvcc q2.cu -o q2  
STUDENT@MIT-ICT-LAB5-25:~/230905213/week7$ ./q2  
Enter a word ABC  
Final str: ABCABA  
STUDENT@MIT-ICT-LAB5-25:~/230905213/week7$ ./q2  
Enter a word HELLO  
Final str: HELLOHELLHELHEH
```